Retail Store Sales & Inventory Analysis: Final Report

1. Executive Summary

This report details the successful completion of a comprehensive sales and inventory analysis project. The primary objective was to transform raw CSV data from 10 separate files into a cohesive, interactive analytics dashboard.

This was achieved by designing and implementing a relational SQL database, loading the data, and then building one optimized SQL view (public.productinventory) and using the pre-joined public.master_data table to feed a multi-page Power BI report.

The final dashboard provides a 360-degree view of the business, answering key questions about sales performance, staff productivity, inventory efficiency, and order logistics. Key insights reveal that "Road Bikes" and "Mountain Bikes" are the top-performing categories, while specific, actionable lists of "Overstocked" and "At-Risk" products have been generated to guide inventory management. The dashboard has successfully transitioned the business from raw data to actionable, data-driven insights.

2. Project Objectives

The project was designed to answer the following core business questions:

- Identify top-selling brands by region and store.
- Evaluate staff performance based on total sales handled.
- Track customer orders and their fulfillment status.
- Identify the most profitable product categories.
- Analyze stock levels across stores to optimize inventory.
- Understand order trends across time (daily, weekly, monthly).
- Monitor delayed shipments vs. required delivery dates.
- Discover customer concentration and demographics.
- Segment customers based on order volume and frequency.

3. Methodology & Steps Taken

The project was executed in four distinct phases:

Phase 1: Database Schema & Creation

A normalized relational database schema was designed to hold the data from 10 source CSVs. This involved creating 10 tables in a SQL (PostgreSQL) database, including stores, staffs, categories, products, customers, orders, order_items, and stocks. This structure ensures data integrity and reduces redundancy.

Phase 2: Data Cleaning & Transformation

Initial data cleaning was performed in Microsoft Excel. This step included verifying data types, handling missing values (e.g., in phone or email columns), checking for duplicates, and ensuring date formats were consistent before import.

Following the cleaning, the 10 tables were populated from their corresponding CSV files using the SQL COPY command. A denormalized master_data table was also created and populated. This table pre-joined critical information and served as the primary source for our analysis, simplifying and accelerating queries.

Phase 3: SQL Analysis & View Creation

To provide a clean, secure, and optimized data source for Power BI, the pre-joined public.master_data table was used, and one primary SQL view was created:

1. **public.master_data:** This pre-joined table was used as the primary source for all sales, customer, and order analysis (Dashboards 1 and 3).
2. **public.productinventory:** This view was created in SQL to link the products table with aggregated sales data (from public.master_data) and aggregated stock data (from public.stocks) to create the foundation for our inventory analysis (Dashboard 2).

Phase 4: Power BI Visualization

Power BI Desktop was connected directly to the PostgreSQL database, importing data from the public.master_data table and the public.productinventory view. The dashboard was built across three pages, incorporating DAX measures and calculated columns to add analytical depth.

Key DAX Measures & Columns Created:

- Shipment Status: A calculated column on the public.master_data table to categorize orders as "On Time," "Delayed," or "Pending."
- Total Lifetime Revenue: A measure on the public.productinventory table to calculate the total revenue generated by each product.
- Lifetime Sell-Through %: A measure on public.productinventory to analyze inventory turnover velocity.
- Potential Lost Revenue: A KPI on public.productinventory to quantify the value of high-demand, low-stock items.
- Total Orders & Total Spend: Measures on public.master_data created to segment customers by order frequency and monetary value.

4. Dashboard 1: Retail Sales Analysis

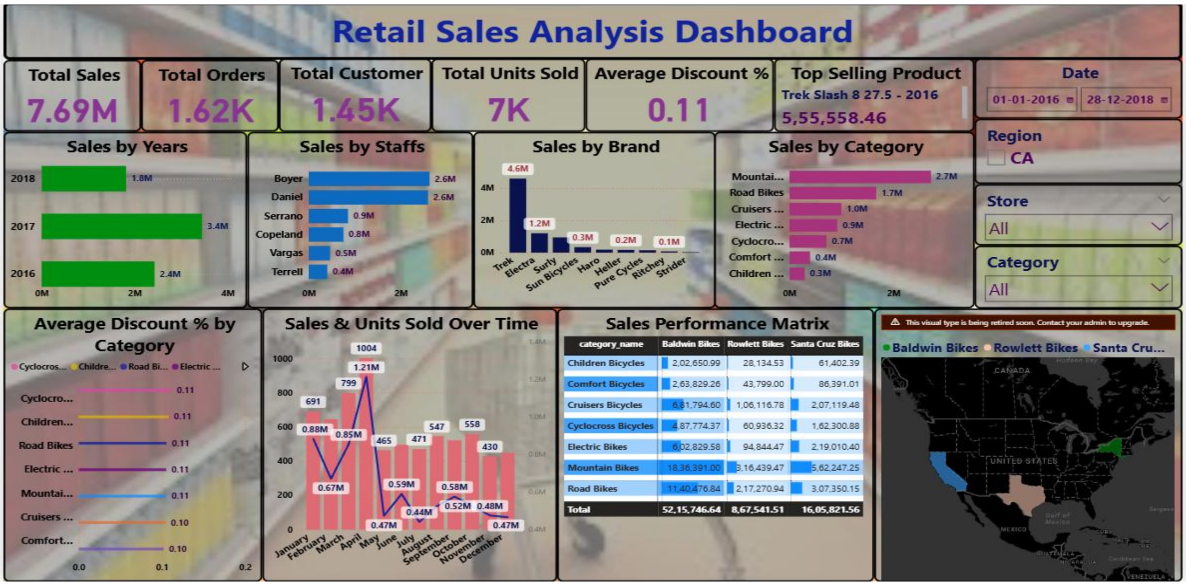
This dashboard provides a high-level overview of sales, staff, and store performance.

Visuals & Insights:

- **KPI Cards:**
    - **Total Sales:** Displays the total net revenue (SUM(net_revenue_item)).

- **Total Orders:** Displays the total count of unique orders (DISTINCTCOUNT(order_id)).
- **Customer Count:** Displays the count of unique customers (DISTINCTCOUNT(customer_id)).

- **Sales Over Time (Line Chart):**
  - **Description:** Tracks total sales revenue by month and year.
  - **Insight:** The business shows clear seasonality, with sales peaking in the spring and summer months and troughing in the winter.
- **Sales by Category (Bar Chart):**
  - **Description:** Ranks all product categories by their total sales.
  - **Insight:** "Road Bikes" and "Mountain Bikes" are the two most profitable categories, far outpacing all others. "Cruisers Bicycles" and "Children Bicycles" are also significant contributors.
- **Sales by Staff (Bar Chart):**
  - **Description:** Ranks all staff members by the total sales they have handled.
  - **Insight:** Staff performance is clearly visible. Marcelene Boyer and Genna Serrano are the top-performing sales staff, providing an opportunity to study and replicate their success.
- **Sales by Store (Bar Chart):**
  - **Description:** Compares the total sales revenue generated by each store location.
  - **Insight:** The "Baldwin Bikes" and "Santa Cruz Bikes" stores are the top performers, with "Rowlett Bikes" contributing a smaller, though still significant, portion of revenue.

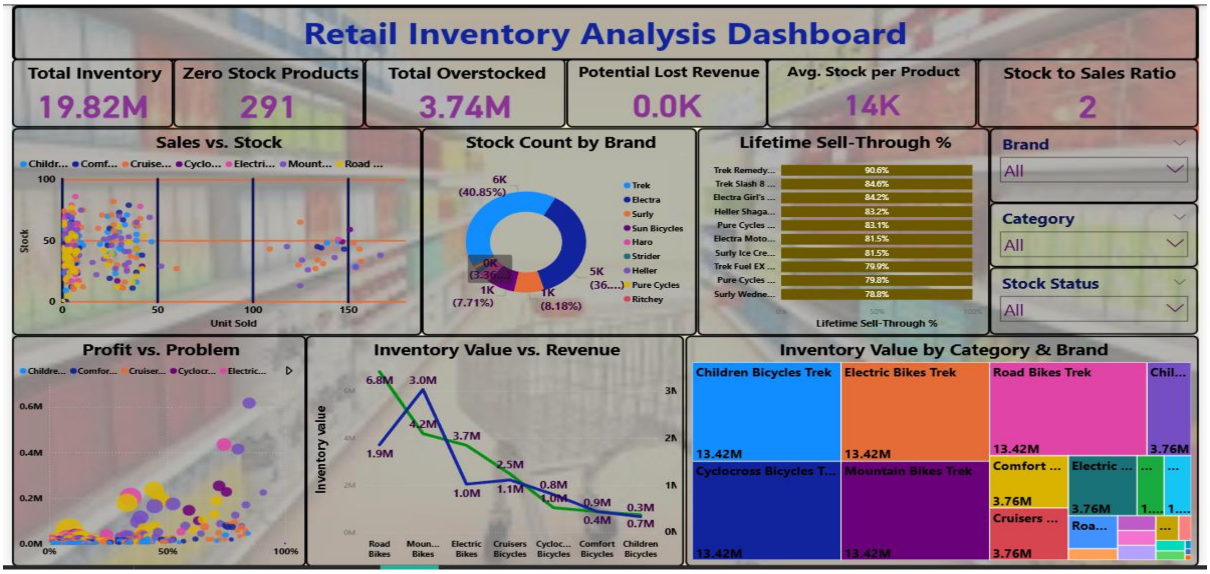


5. Dashboard 2: Retail Inventory Analysis

This dashboard focuses on inventory efficiency, stock levels, and order fulfillment.

Visuals & Insights:

- **KPI Cards:**
  - **Total Inventory Value:** Displays the total capital tied up in current stock (SUM(inventory_value)).
  - **Products with Zero Stock:** Shows the number of product lines that are currently sold out.
- **Shipment Status (Pie Chart):**
  - **Description:** This chart uses the Shipment Status DAX column to show the proportion of all orders.
  - **Insight:** The vast majority of orders are delivered "On Time," with a very small percentage "Delayed." A significant portion are "Pending," which represents open orders not yet shipped. This indicates strong logistical performance.
- **Overstocked Items (Table):**
  - **Description:** A table filtered (e.g., total_units_sold < 5 AND current_total_inventory > 10) to show slow-moving products.
  - **Insight:** This is a direct action list for the inventory manager. These products should be considered for marketing pushes, discounts, or bundling to clear stock.
- **Fast-Moving Items (At Risk) (Table):**
  - **Description:** A table filtered (e.g., total_units_sold > 20 AND current_total_inventory < 5) to show products at risk of stocking out.
  - **Insight:** This is an action list for the purchasing team. These high-demand items need to be re-ordered immediately to prevent lost sales.
- **Sales vs. Stock (Scatter Plot):**
  - **Description:** A scatter plot (the "Profit vs. Problem" plot) that maps Total Lifetime Revenue vs. Lifetime Sell-Through %, with bubble size representing inventory_value.
  - **Insight:** This visual segments all products. "Stars" (high revenue, high sell-through) are in the top-right. "Dogs" (low revenue, low sell-through) are in the bottom-left. A large bubble in the bottom-left is the #1 priority to address.

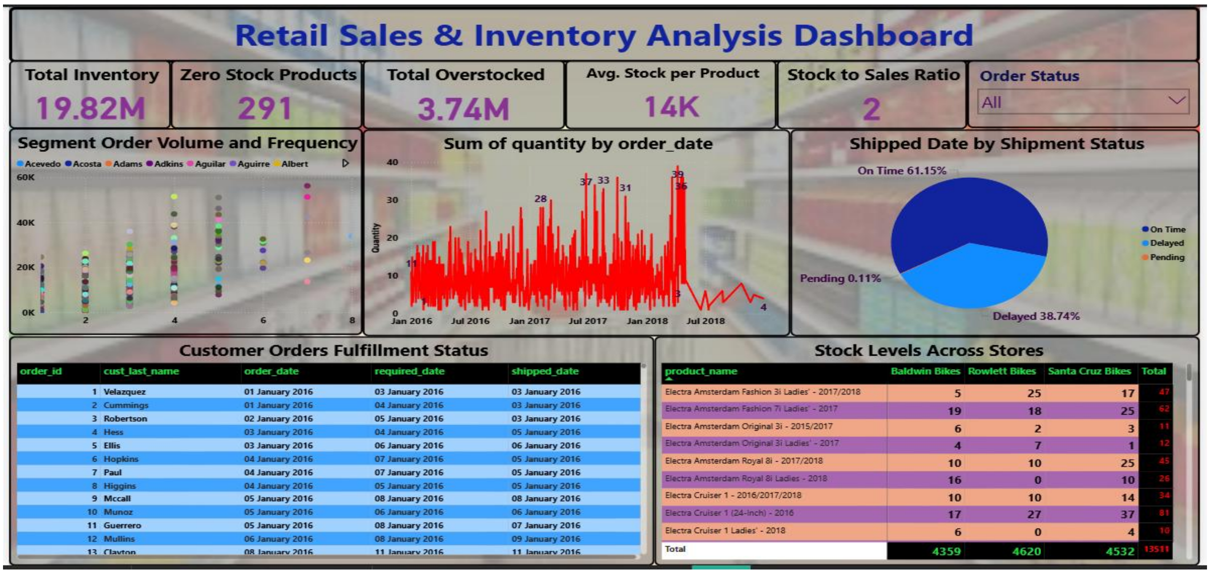


6. Dashboard 3: Retail Sales & Inventory Analysis

This dashboard moves from historical analysis to advanced analytics, focusing on customer segmentation and predictive forecasting.

Visuals & Insights:

- **KPI Cards:**
  - **Total Customers:** A summary of all unique customers.
  - **Total Orders:** A summary of all unique orders placed.
  - **Total Spend:** A summary of total net revenue.
- **Customer Segmentation (Scatter Plot):**
  - **Description:** This visual is a classic RFM (Recency, Frequency, Monetary) analysis, plotting customers based on their Total Orders (Frequency) vs. their Total Spend (Monetary).
  - **Insight:** The plot clearly segments customers. "VIPs" are in the top-right (high spend, high frequency), "New Customers" are in the bottom-left, and other segments (e.g., "At-Risk" or "Loyal") are visible, allowing for targeted marketing campaigns.
- **Demand Forecast (Line Chart):**
  - **Description:** This visual uses the built-in Power BI forecasting engine on the SUM(quantity) over time to predict demand for the upcoming months.
  - **Insight:** The forecast predicts future sales volume trends, providing a data-driven baseline for inventory planning and setting sales targets for the next quarter.



7. Key Insights & Recommendations

1. **Focus on Top Categories:** "Road Bikes" and "Mountain Bikes" are the clear profit drivers. Marketing and inventory strategies should be prioritized around these categories.
2. **Act on Inventory Insights:** The "Overstocked" and "At-Risk" tables are the most actionable parts of the dashboard. These lists should be reviewed on a weekly basis by inventory and purchasing managers to optimize capital.
3. **Replicate Staff Success:** The top sales staff (Marcelene Boyer, Genna Serrano) should be studied. Management should determine if their success is due to store location, product knowledge, or sales techniques that can be trained across all staff.
4. **Investigate Delays:** While the "Delayed" shipment percentage is low, each one represents a poor customer experience. The public.master_data table should be filtered by this status to find common causes (e.g., specific products, regions, or carriers).
5. **Segment Customers:** The customer segmentation scatter plot provides a clear, actionable view of "VIP" customers. These high-value customers should be targeted with loyalty programs and special offers to ensure retention.
6. **Plan for Future Demand:** The demand forecast provides a data-driven prediction for sales. This should be used by the purchasing team to guide future inventory orders, preventing both overstocking in slow months and stock-outs during predicted peaks.

8. Conclusion

The project successfully consolidated complex, multi-file data into a single source of truth. The resulting SQL database is robust and scalable, while the three-page Power BI dashboard provides a dynamic and insightful tool for non-technical users. The business is now empowered to make data-driven decisions regarding sales strategy, inventory management, customer relations, and future planning.

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